1 **(a)** kg m s^{-2} B1

(b) kg m^{-1} s^{-1} B1

(c) **(i)** $v^2 = 2gs$
$= 2 \times 9.8 \times 4.5$ C1
$v = 9.4$ m s^{-1} A1 **[2]**

(ii) *either*
F (= $3.2 \times 10^{-4} \times 1.2 \times 10^{-2} \times 9.4$) = 3.6×10^{-5} N M1
weight of sphere (= mg = $15 \times 10^{-3} \times 9.8$) = 0.15 N M1
$3.6 \times 10^{-5} << 0.15$, so justified A1 **[3]**
or
$mg = crv_T$ (M1)
terminal speed = 3.8×10^4 m s^{-1} (M1)
$9.4 << 3.8 \times 10^4$, so justified (A1)

2 **(a)** **(i)** point at which whole weight of body M1